Natural deduction proof editor and checker

This is a demo of a proof checker for Fitch-style natural deduction systems found in many popular introductory logic textbooks. The specific system used here is the one found in ***forall x: Calgary***. (Although based on ***forall x: an Introduction to Formal Logic***, the proof system in that original version differs from the one used here and in *forall x: Calgary*. However, the system also supports the rules used in the ***forall x: Cambridge*** remix.)

Create a new problem

Select if TFL or FOL syntax:
TFL FOL

Premises (separate with "," or ";"):

Conclusion:

CREATE PROBLEM

Sample exercise sets

- Sample Truth-Functional Logic exercises (Chap. 15, ex. C; Chap. 17, ex. B)
- Sample First-Order Logic exercises (Chap. 32, ex. E; Chap. 34, ex. A)

Instructions

TFL atomic sentences: (single uppercase letters)	`A, B, X, etc.`
FOL atomic sentences: (single uppercase letters *other than A or E* followed by	`Pa, Fcdc, a = d, etc.`

Rules

Basic rules

m	$\mathscr{A}$	
	$\mathscr{A}$	R m

m	$\mathscr{A}$	
n	$\mathscr{B}$	
	$\mathscr{A} \wedge \mathscr{B}$	$\wedge$I m, n

m	$\mathscr{A} \wedge \mathscr{B}$	
	$\mathscr{A}$	$\wedge$E m

m	$\mathscr{A} \wedge \mathscr{B}$	
	$\mathscr{B}$	$\wedge$E m

m	$\mathscr{A}$	
	$\mathscr{A} \vee \mathscr{B}$	$\vee$I m

m	$\mathscr{A}$	
	$\mathscr{B} \vee \mathscr{A}$	$\vee$I m

lowercase letters a–w without parentheses, or identities)	
For negation you may use any of the symbols:	¬ ~ ~ − –
For conjunction you may use any of the symbols:	∧ ^ & . · *
For disjunction you may use any of the symbols:	∨ v
For the biconditional you may use any of the symbols:	↔ ≡ <–> <> (or in TFL only: =)
For the conditional you may use any of the symbols:	→ ⇒ ⊃ –> >
For the universal quantifier (FOL only), you may use any of the symbols:	∀x (∀x) Ax (Ax) (x) ⋀x
For the existential quantifier (FOL only), you may use any of the symbols:	∃x (∃x) Ex (Ex) ⋁x
For a contradiction you may use any of the symbols:	⊥ XX #

The following buttons do the following things:

	= delete this line
	= add a line below this one
	= add a new subproof below this line
	= add a new line below this subproof to the parent subproof
	= add a new subproof below this subproof to the parent subproof

m	$\mathscr{A} \vee \mathscr{B}$	
i	│ $\mathscr{A}$	
j	│ $\mathscr{C}$	
k	│ $\mathscr{B}$	
l	│ $\mathscr{C}$	
	$\mathscr{C}$	$\vee$E m, i–j, k–l

i	│ $\mathscr{A}$	
j	│ $\mathscr{B}$	
	$\mathscr{A} \rightarrow \mathscr{B}$	$\rightarrow$I i–j

m	$\mathscr{A} \rightarrow \mathscr{B}$	
n	$\mathscr{A}$	
	$\mathscr{B}$	$\rightarrow$E m, n

i	│ $\mathscr{A}$	
j	│ $\mathscr{B}$	
k	│ $\mathscr{B}$	
l	│ $\mathscr{A}$	
	$\mathscr{A} \leftrightarrow \mathscr{B}$	$\leftrightarrow$I i–j, k–l

m	$\mathscr{A} \leftrightarrow \mathscr{B}$	
n	$\mathscr{A}$	
	$\mathscr{B}$	$\leftrightarrow$E m, n

m	$\mathscr{A} \leftrightarrow \mathscr{B}$	
n	$\mathscr{B}$	
	$\mathscr{A}$	$\leftrightarrow$E m, n

i	│ $\mathscr{A}$	
j	│ $\bot$	
	$\neg\mathscr{A}$	$\neg$I i–j

m	$\neg\mathscr{A}$	
n	$\mathscr{A}$	
	$\bot$	$\neg$E m, n

Apart from premises and assumptions, each line has a cell immediately to its right for entering the justifcation. Click on it to enter the justification as, e.g. “&I 1,2”.

Hopefully it is otherwise more or less obvious how to use it.

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m	$\bot$	
	$\mathscr{A}$	X m

i	$\neg\mathscr{A}$	
j	$\bot$	
	$\mathscr{A}$	IP i–j

m	$\mathscr{A}(\ldots c \ldots c \ldots)$	
	$\forall x \mathscr{A}(\ldots x \ldots x \ldots)$	∀I m

m	$\forall x \mathscr{A}(\ldots x \ldots x \ldots)$	
	$\mathscr{A}(\ldots c \ldots c \ldots)$	∀E m

m	$\mathscr{A}(\ldots c \ldots c \ldots)$	
	$\exists x \mathscr{A}(\ldots x \ldots c \ldots)$	∃I m

m	$\exists x \mathscr{A}(\ldots x \ldots x \ldots)$	
i	$\mathscr{A}(\ldots c \ldots c \ldots)$	
j	$\mathscr{B}$	
	$\mathscr{B}$	∃E m,

	$c = c$	=I

m	$a = b$	
n	$\mathscr{A}(\ldots a \ldots a \ldots)$	
	$\mathscr{A}(\ldots b \ldots a \ldots)$	=E m, n

m	$a = b$	
n	$\mathscr{A}(\ldots b \ldots b \ldots)$	
	$\mathscr{A}(\ldots a \ldots b \ldots)$	=E m, n

Derived rules

m	$\mathscr{A} \vee \mathscr{B}$	
n	$\neg\mathscr{A}$	
	$\mathscr{B}$	DS m, n

m	$\mathscr{A} \vee \mathscr{B}$	
n	$\neg\mathscr{B}$	
	$\mathscr{A}$	DS m, n

m	$\mathscr{A} \rightarrow \mathscr{B}$	
n	$\neg\mathscr{B}$	
	$\neg\mathscr{A}$	MT m, n

m	$\neg\neg\mathscr{A}$	
	$\mathscr{A}$	DNE m

i	| $\mathscr{A}$	
j	| $\mathscr{B}$	
k	| $\neg\mathscr{A}$	
l	| $\mathscr{B}$	
	$\mathscr{B}$	LEM i–j, k–l

$$\begin{array}{l|ll} m & \neg(\mathscr{A} \lor \mathscr{B}) & \\ & \neg\mathscr{A} \land \neg\mathscr{B} & \text{DeM } m \end{array}$$

$$\begin{array}{l|ll} m & \neg\mathscr{A} \land \neg\mathscr{B} & \\ & \neg(\mathscr{A} \lor \mathscr{B}) & \text{DeM } m \end{array}$$

$$\begin{array}{l|ll} m & \neg(\mathscr{A} \land \mathscr{B}) & \\ & \neg\mathscr{A} \lor \neg\mathscr{B} & \text{DeM } m \end{array}$$

$$\begin{array}{l|ll} m & \neg\mathscr{A} \lor \neg\mathscr{B} & \\ & \neg(\mathscr{A} \land \mathscr{B}) & \text{DeM } m \end{array}$$

$$\begin{array}{l|ll} m & \forall x \neg\mathscr{A} & \\ & \neg\exists x \mathscr{A} & \text{CQ } m \end{array}$$

$$\begin{array}{l|ll} m & \neg\exists x \mathscr{A} & \\ & \forall x \neg\mathscr{A} & \text{CQ } m \end{array}$$

$$\begin{array}{l|ll} m & \exists x \neg\mathscr{A} & \\ & \neg\forall x \mathscr{A} & \text{CQ } m \end{array}$$

$$\begin{array}{l|ll} m & \neg\forall x \mathscr{A} & \\ & \exists x \neg\mathscr{A} & \text{CQ } m \end{array}$$

Rules for Cambridge

$$\begin{array}{l|ll} m & \mathscr{A} & \\ n & \neg\mathscr{A} & \\ & \bot & \bot\text{I } m, n \end{array}$$

$$\begin{array}{l|ll} m & \bot & \\ & \mathscr{A} & \bot\text{E } m \end{array}$$

$$\begin{array}{l|ll} i & \quad \mathscr{A} & \\ j & \quad \mathscr{B} & \\ k & \quad \neg\mathscr{A} & \\ l & \quad \mathscr{B} & \\ & \mathscr{B} & \text{TND } i\text{–}j,\ k\text{–}l \end{array}$$